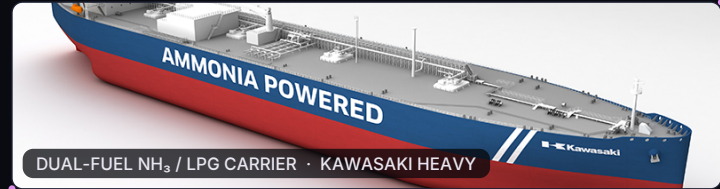
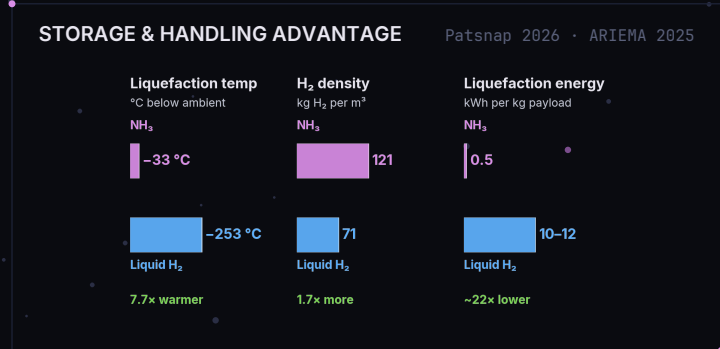
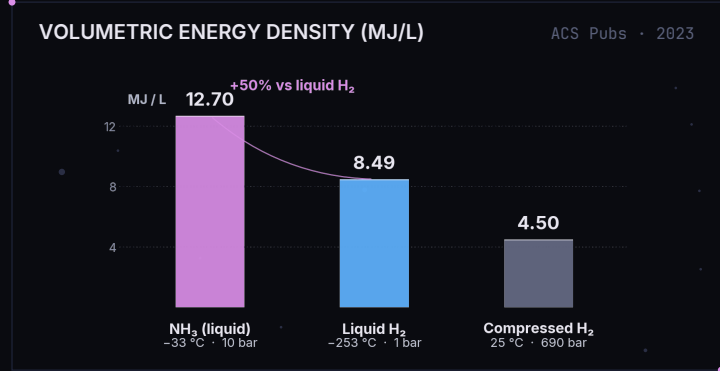


[NH₃ · ENERGY DENSITY NODE · VERIFIED]

github.com/StewAlexander-com/global-tensor-network
addendum 01 · one page · cited

Ammonia is a high-density H₂ carrier.

A one-page evidence brief for the NH₃ node in the Global Systems Tensor Network. Figures match the repo source and cited primary literature.



WHY NH₃ BELONGS ON THE ENERGY-DENSITY NODE

Energy density is the #1 systemic lever in the network (12 connections, more than any other node). Ammonia raises it without trading climate, agriculture, or supply-chain resilience. At 12.7 MJ/L liquid (-33 °C, ~10 bar), NH₃ carries 50% more energy per litre than liquid H₂. Synthesizable from water + air + electricity — the structural fix the model points to.

MORE ENERGY PER LITRE
12.7 MJ / L vs 8.49 for liquid H₂
Liquid NH₃ stores 50% more chemical energy per litre than liquid H₂, and 1.7× more H₂ per cubic metre (121 vs 71 kg-H₂/m³).
ACS Publications, 2023

01

EASIER TO LIQUEFY & SHIP
-33 °C vs -253 °C for LH₂
NH₃ liquefies 220 °C warmer than H₂ — cutting liquefaction energy from ~10-12 to ~0.5 kWh per kg of H₂ payload.
Patsnap, 2026 · ARIEMA, 2025

02

INFRASTRUCTURE EXISTS
180 Mt / yr produced via Haber-Bosch
~200 ammonia carriers and 150+ ports already handle NH₃ at scale. Synthesizable from water + air + electricity — routes around fossil chokepoints.
Heise / IFA, 2024 · ARIEMA, 2025

03



SOURCES

- ¹ ACS Publications, Advances and Perspectives of H₂ Production from NH₃, Energy & Fuels (2023). NH₃ 12.7 MJ/L; LH₂ 8.49 MJ/L; NH₃ storage 9.9 bar / 25 °C. pubs.acs.org
- ² Patsnap, Ammonia as hydrogen carrier: 2026 technology landscape (2026). 121 vs 71 kg-H₂/m³; liquefaction 0.5 vs 10-12 kWh/kg. patsnap.com
- ³ Heise / IFA: ~180 Mt/yr NH₃ via Haber-Bosch (2024). heise.de
- ⁴ ARIEMA (2025): 150+ ports handle NH₃ today. ariema.com